INCH-POUND
MIL-DTL-49285/16A
31 March 2010
SUPERSEDING
MIL-C-49285/16
19 October 1989

DETAIL SPECIFICATION SHEET

# CABLE, SPECIAL PURPOSE, ELECTRICAL, THREE INDIVIDUALLY SHIELDED PAIRS OF 22 AWG SOLID WIRE

Inactive for new design after 16 June 1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-49285.

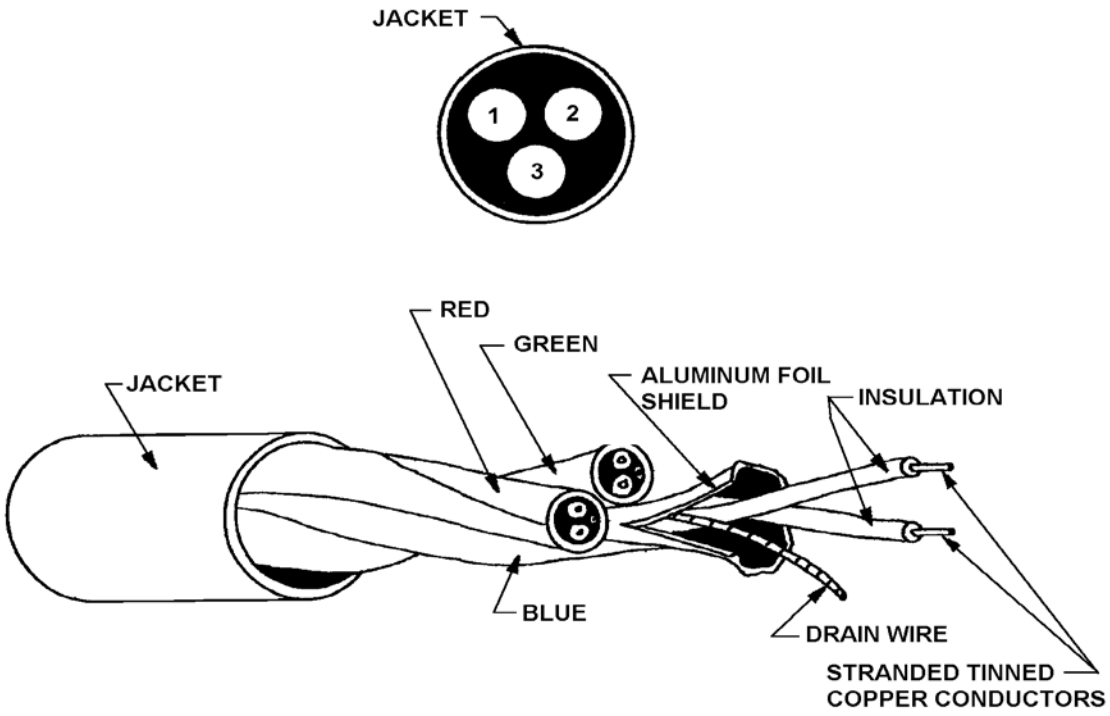



FIGURE 1. Cable illustration.

AMSC N/A

FSC 6145

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Engineering information: See table I.

Note: Not for aerospace use.

TABLE I. Description, electrical.

| Electrical characteristics | Requirements |
|---|---|
| Nominal capacitance between conductors | 40 pF/ft (picofarads/foot) |
| Nominal capacitance between the conductor and the shield with one conductor tied to the shield | 77 pF/ft |
| Continuous working voltage | 300 V rms maximum between conductors |
| Nominal capacitance between adjacent shields | 150 pF/ft |
| Continuous working voltage between adjacent shields | 50 volts maximum |
| Shield dc resistance | 14 ohms/1,000 feet maximum |

TABLE II. Description, physical.

| Components | Construction details |
|---|---|
| Number of pairs | Three individually shielded |
| Conductor type and wire size | Solid tinned copper 22 AWG |
| Conductor insulation | Polyvinyl chloride |
| Conductor insulation thickness | 0.013 inch nominal thickness |
| Drain wire type and size | Solid tinned copper 22 AWG |
| Jacket material | Polyvinyl chloride |
| Jacket thickness | 0.037 inch |
| Finished cable diameter | 0.293 inch maximum outside diameter |
| Cable style (UL) | 2464 |
| Tensile strength (jacket) | 2,000 pounds per square inch minimum |
| Elongation (jacket) | 150 percent minimum |
| Overall cabling lay length | 2.0 twists per foot ± 10 percent |

Cabled pair orientation. The position of the red and green pair shield shall determine the rotation direction for pair location (see figure 1).



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TABLE III. Shield color code.

| Pair numbers | Shield color |
|---|---|
| 1st pair | Red |
| 2nd pair | Green |
| 3rd pair | Blue |

REQUIREMENTS:

Design and construction: (see table II and table III).

Shield location and orientation. The polyester aluminum shield is to be located on the outer circumference of the pair, with the aluminum foil side outward. Insulation on the interior of the shield is to be complete with no aluminum available for contact.

Drain wire location. The drain wire is to be spirally located between the noise reducing tape and the aluminum foil shield, and is to be in continuous contact with the aluminum surface of the foil shield and the noise reducing tape throughout the cable.

Cable temperature rating. The cable temperature rating shall be -20°C to +80°C.

Flammability. The cable shall pass UL 2556 flame test requirements.

Shield integrity test.*

| Frequency | Response ** |
|---|---|
| 100 kHz | 114 dB below reference |
| 500 kHz | 105 dB below reference |
| 1 MHz | 99 dB below reference |
| 5 MHz | 85 dB below reference |
| 10 MHz | 79 dB below reference |
| 15 MHz | 74 dB below reference |
| 20 MHz | 70 dB below reference |
| 25 MHz | 65 dB below reference |
| 30 MHz | 61 dB below reference |
| 100 MHz | 42 dB below reference |

* Any two readings may deviate to a value of 0.9 times the stated limits.

** All values are a minimum numerical value.

Fixture resonance will occur between 40 and 70 MHz.

Crosstalk test limits. All values are minimums.

| Frequency (kHz) | Near end crosstalk (dB) | Far end crosstalk (dB) |
|---|---|---|
| 40 | 58 | 58 |
| 70 | 56 | 55 |
| 100 | 54 | 53 |
| 1,000 | 55 | 52 |

Insulation resistance between shields: 100 megohms/1,000 feet minimum.



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Durometer hardness. The cable jacket shall have a “Shore A” hardness of 82 ± 5.

Part or Identifying Number (PIN): The PIN shall be M49285/16.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. This document references MIL-DTL-49285 and UL 2556.

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

Review activities:
Army – MI
Air Force – 99
DLA - IS

(Project 6145-2009-020)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change. You should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.



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